

AV2 Codec ISO Media File Format Binding

Working Group Draft, 15 September 2026



This version:

<https://aomediacodec.github.io/av2-isobmff/v1.0.0.html>

Issue Tracking:

[GitHub](#)

Editors:

TBD

TBD

Warning

This specification is still at draft stage and should not be referenced other than as a working draft.

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Abstract

This document specifies the storage format for [AV2] bitstreams in [ISOBMFF] tracks as well as in [CMAF] files.

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§ 1. Terminology

The key words "**MUST**", "**MUST NOT**", "**REQUIRED**", "**SHALL**", "**SHALL NOT**", "**SHOULD**", "**SHOULD NOT**", "**RECOMMENDED**", "**NOT RECOMMENDED**", "**MAY**", and "**OPTIONAL**" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

All of the text of this specification is normative except sections explicitly marked as non-normative, examples, and notes.

Examples in this specification are introduced with the words "for example" or are set apart from the normative text with `class="example"`, like this:

EXAMPLE 1

This is an example of an informative example.

Informative notes begin with the word "Note" and are set apart from the normative text with `class="note"`, like this:

NOTE: This is an example of an informative note.

§ 2. Bitstream features overview

An [AV2 bitstream](#) is composed of a sequence of [OBUs](#) (Open Bitstream Units), grouped into [Temporal Units](#). A Temporal Unit consists of a [Temporal Delimiter OBU](#) and all following OBUs up to, but not including, the next Temporal Delimiter OBU. Each Temporal Unit is associated with a single presentation time.

Every OBU begins with a 1- or 2-byte header that carries the [obu_type](#) and the temporal layer identifier [obu_tlayer_id](#). When the OBU header extension is present (i.e. when [obu_header_extension_flag](#) is equal to 1 in that header), two additional layer identifiers are carried: the embedded layer identifier [obu_mlayer_id](#) and the extended layer identifier [obu_xlayer_id](#). Depending on its type, an OBU can carry configuration information (e.g., a [Sequence Header OBU](#) or [Layer Configuration Record OBU](#)), metadata, or coded video data.

A Temporal Unit **MAY** contain multiple coded pictures but only one is output. AV2 defines a number of coded-frame OBU types, including [Closed Loop Key OBUs](#), [Open Loop Key OBUs](#), regular and leading tile-group OBUs, [Switch OBUs](#), [OBU_RAS_FRAME](#), [Bridge Frame OBUs](#), and TIP (Temporally Interpolated Prediction) frames.

Frames that can be decoded without reference to other frames are [Key Frames](#) (carried in a [Closed Loop Key OBU](#) or an [Open Loop Key OBU](#)) and [Intra frames](#). All other coded frames - including [Inter Frames](#), [Switch Frames](#), RAS frames, Bridge frames, TIP frames, and frames with [show_existing_frame](#) equal to 1 - have coding dependencies on other frames in the same or previous Temporal Units.

A [Closed Random Access](#) is the random access process that applies to an extended layer whose first coded frame is a [Key Frame](#) carried in a [Closed Loop Key OBU](#). It starts a new coded video sequence for the extended layer; the layer's reference frame buffers are invalidated and every subsequent frame of that layer can be decoded. This is the primary random-access mechanism.

An [Open Random Access](#) is the random access process that applies to an extended layer whose first coded frame is a [Key Frame](#) carried in an [Open Loop Key OBU](#). During sequential decoding, an [Open Random Access](#) point does not start a new coded video sequence; however, when a decoder initiates decoding at that point, the decoding process is treated as if it were the start of a new coded video sequence for the extended layer. [Leading frames](#) that follow an [Open Random Access](#) point **MAY** reference frames that precede it, and are discarded when decoding begins at the [Open Random Access](#) point.

In addition to Key Frames, AV2 provides two further mechanisms that support random access or stream switching: a RAS frame (OBU_RAS_FRAME) is an inter-predicted frame that uses long-term reference frames and enables random access or switching without inserting a full key frame; a [Switch Frame](#) (OBU_SWITCH) is an inter-predicted frame that enables switching between representations at a given point, typically in adaptive streaming scenarios.

AV2 supports three scalability dimensions: temporal layers (identified by [obu_tlayer_id](#)), embedded layers (identified by [obu_mlayer_id](#)), and extended layers (identified by [obu_xlayer_id](#)). The reserved [obu_xlayer_id](#) value GLOBAL_XLAYER_ID (equal to 31, as defined in [\[AV2\]](#)) denotes global scope and is used by OBUs that do not belong to any specific extended layer (such as the [Temporal Delimiter OBU](#)). A bitstream that uses a single distinct non-global value of [obu_xlayer_id](#) is a [Singlestream](#); a bitstream that uses two or more distinct non-global values is a [Multistream](#).

AV2 defines two distinct [Metadata OBU](#) types for carrying metadata: OBU_METADATA_SHORT, using the [metadata short OBU syntax](#), and OBU_METADATA_GROUP, using the [metadata group OBU syntax](#). A separate OBU type carries film grain synthesis parameters; its presence is indicated by the [film_grain_params_present](#) flag in the [Sequence Header OBU](#).

The [Sequence Header OBU](#) also carries the bitstream's profile, level, and - when applicable - tier, as defined in Annex A of [\[AV2\]](#).

The [Sequence Header OBU](#) carries a [monotonic_output_order_flag](#) that defines the output mode of the associated coded video sequence.

When [monotonic_output_order_flag](#) is equal to 1, the output order of coded output frame equals their decoding order within the coded video sequence. Within a sample, the composition time (CTS) of the associated output frame is equal to the sample's decoding time (DTS); the sample **MAY** additionally contain hidden frames.

When [monotonic_output_order_flag](#) is equal to 0, the output order of coded output frames **MAY** differ from their decoding order. The composition time of each sample is then derived from a ['ctts](#) box that signals the offset between the sample's decoding time (DTS) and the composition time of its associated output frame. As in the monotonic case, a sample **MAY** additionally contain hidden frames.

NOTE: Non-monotonic output mode is new in AV2; AV1 supports only the monotonic mode.

§ 3. Basic Encapsulation Scheme

This section describes the basic data structures used to signal encapsulation of [AV2 bitstreams](#) in [\[ISOBMFF\]](#) containers.

§ 3.1. General Requirements & Brands

A file conformant to this specification satisfies the following:

- It SHALL conform to the normative requirements of [ISOBMFF]
- It SHALL have the ‘**av02**’ brand among the compatible brands array of the FileTypeBox
- It SHALL contain at least one track using an [AV2SampleEntry](#), possibly transformed by encryption as specified in § 5 Common Encryption
- It SHOULD indicate a structural ISOBMFF brand among the compatible brands array of the FileTypeBox, such as ‘[iso6](#)’

§ 3.2. AV2 Sample Entry and Configuration

§ 3.2.1. AV2 Sample Entry

The **AV2SampleEntry** extends [VisualSampleEntry](#) and signals a track carrying AV2 coded video samples.

```
class AV2SampleEntry extends VisualSampleEntry('av02') {  
    AV2CodecConfigurationBox config;  
}
```

An [AV2SampleEntry](#) **SHALL** contain exactly one [AV2CodecConfigurationBox](#).

An [AV2SampleEntry](#) **SHALL** contain a ‘[colr](#)’ box with the ‘[nclx](#)’ colour type. Each value among `colour_primaries`, `transfer_characteristics`, `matrix_coefficients`, and `full_range_flag` carried in that ‘[colr](#)’ box:

- **SHALL** equal the corresponding effective value of the AV2 bitstream carried by the track if the latter is anything but 2 (Unspecified);
- **SHALL** take precedence over the corresponding effective value of the AV2 bitstream carried by the track.

NOTE: An [AV2SampleEntry](#) may contain additional sub-boxes permitted by [ISOBMFF] for a [VisualSampleEntry](#), such as ‘[pasp](#)’, ‘[clli](#)’, ‘[mdcv](#)’ or ‘[btrt](#)’.

§ 3.2.2. AV2 Codec Configuration Box

The **AV2CodecConfigurationBox** carries the decoder initialisation data required to decode the AV2 coded video samples of a track.

```
aligned(8) class AV2CodecConfigurationBox extends Box('av2C')  
{  
    unsigned int(8) reserved1 = 0;  
    unsigned int(8) config_obus_count_minus1;  
    for (i = 0; i < config_obus_count_minus1 + 1; ++i) {  
        unsigned int(8) config_obu[];  
    }  
}
```

reserved **SHALL** be set to 0. Readers **SHALL** ignore its value.

`config_obus_count_minus1` is the number of `config_obu` minus one.

`config_obu` is a byte array carrying one OBU and its length as described in [§ 3.2.3 config_obu](#).

§ 3.2.3. **config_obu**

Each `config_obu` **SHALL** carry a single OBU preceded by a `leb128()` `num_bytes_in_obu` value giving the byte length of that OBU, as specified in the [Length delimited bitstream format](#) from Annex B of [\[AV2\]](#).

`config_obu` **SHALL NOT** contain OBUs of the types forbidden there as stated in [Figure 3.1](#).

AV2 OBU type	Be present in AV2CodecConfigurationBox (in config_obu)	Be present in a sample
OBU_SEQUENCE_HEADER	SHALL	SHALL NOT
OBU_TEMPORAL_DELIMITER	SHALL NOT	SHALL NOT (the sample boundary already delimits the temporal unit)
OBU_MULTI_FRAME_HEADER	SHALL NOT	MAY
OBU_CLOSED_LOOP_KEY	SHALL NOT	MAY
OBU_OPEN_LOOP_KEY	SHALL NOT	MAY
OBU_LEADING_TILE_GROUP	SHALL NOT	MAY
OBU_REGULAR_TILE_GROUP	SHALL NOT	MAY
OBU_METADATA_SHORT	MAY	MAY
OBU_METADATA_GROUP	MAY	MAY
OBU_SWITCH	SHALL NOT	MAY
OBU_LEADING_SEF	SHALL NOT	MAY
OBU_REGULAR_SEF	SHALL NOT	MAY
OBU_LEADING_TIP	SHALL NOT	MAY
OBU_REGULAR_TIP	SHALL NOT	MAY
OBU_BUFFER_REMOVAL_TIMING	SHALL NOT	MAY
OBU_LAYER_CONFIGURATION_RECORD	MAY	SHALL NOT
OBU_ATLAS_SEGMENT	SHALL NOT	MAY
OBU_OPERATING_POINT_SET	SHALL NOT	MAY
OBU_BRIDGE_FRAME	SHALL NOT	MAY
OBU_MSDO	SHALL NOT	MAY
OBU_RAS_FRAME	SHALL NOT	MAY
OBU_QUANTIZATION_MATRIX	SHALL NOT	MAY
OBU_FILM_GRAIN	MAY	MAY
OBU_CONTENT_INTERPRETATION	MAY	SHALL NOT
OBU_PADDING	SHALL NOT	SHALL NOT (byte-level padding within a track is achieved using ISOBMFF mechanisms, e.g.

'free'/'skip' boxes or sample size signalling)

Figure 3.1: List of AV2 OBU types and where they can be found

The first config_obu **SHALL** carry a [Sequence Header OBU](#).

The AV2CodecConfigurationBox **MAY** contain additional OBUs whose content is constant for the track's samples, such as static [Metadata OBUs](#).

§ 3.3. AV2 Sample Format

§ 3.3.1. General

Each AV2 sample carries exactly one [Temporal Unit](#) of the [AV2 bitstream](#); the sample boundary delimits the temporal unit.

Each [OBU](#) within a sample **SHALL** be framed using the [Length delimited bitstream format](#) specified in Annex B of [\[AV2\]](#): each OBU is preceded by a `leb128() num_bytes_in_obu` field giving the byte length of the OBU that follows. [Figure 3.2](#) illustrates this mapping.

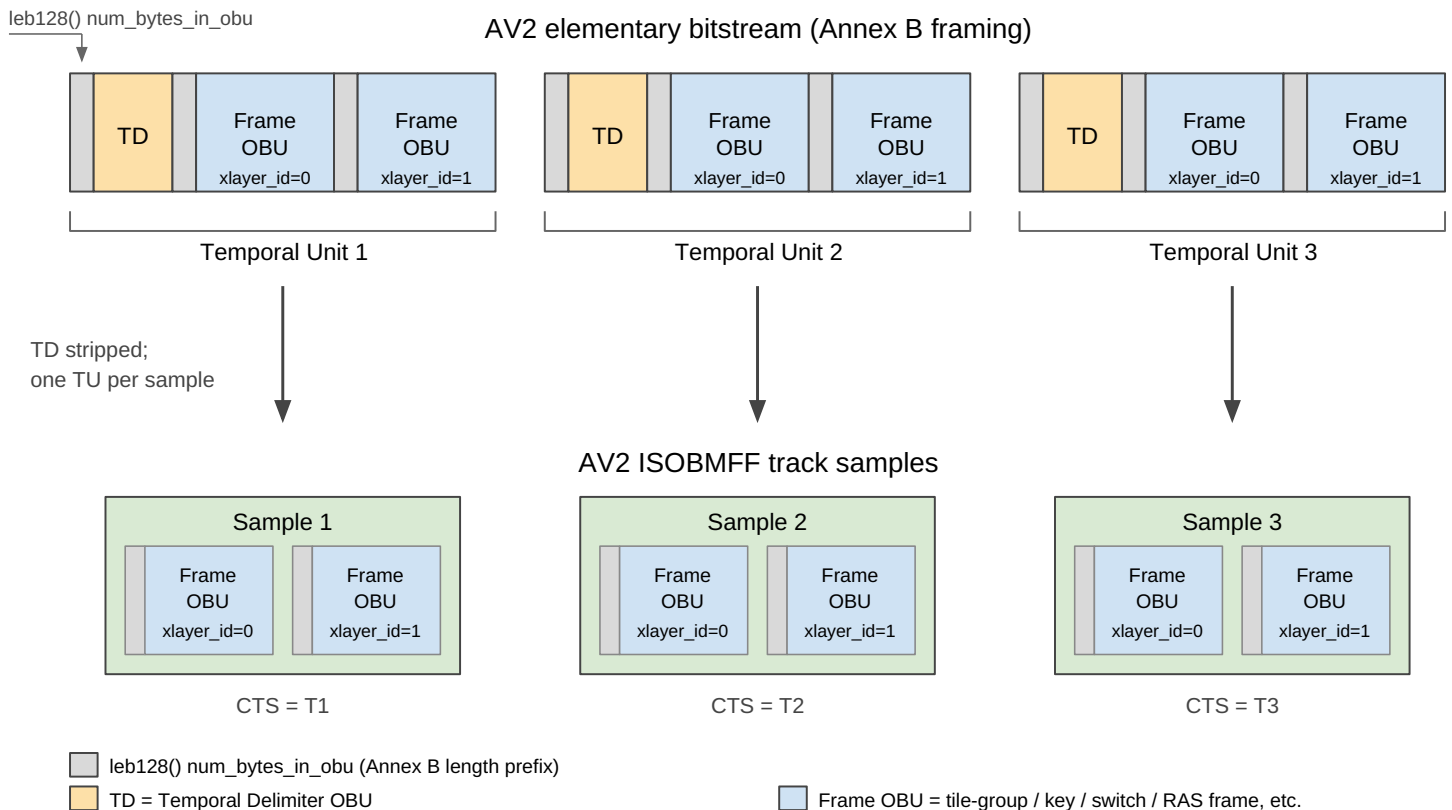


Figure 3.2: Example mapping of AV2 temporal units into the AV2 ISOBMFF sample format.

§ 3.3.2. OBUs in samples

The OBUs within a sample **SHALL** appear in the order specified by the [ordering of OBUs in a temporal unit](#) in [AV2], with the following constraints:

- the OBUs within a sample **SHALL NOT** contain OBUs of the types forbidden there as stated in [Figure 3.1](#);
- a sample **SHOULD NOT** contain any other OBU whose contents do not change for the duration of the active [AV2SampleEntry](#), in order to avoid redundant data; for example, a [Metadata OBU](#) whose contents are constant should be carried in the [AV2CodecConfigurationBox](#) rather than repeated in every sample, or discarded from the samples and conveyed at the ISOBMFF level.

The handling of the [Temporal Delimiter OBU](#) depends on how a track's samples are consumed. When the samples are passed to an AV2 decoder API that already treats each sample as one [Temporal Unit](#), a [Temporal Delimiter OBU](#) is not required. When the samples are reconstructed into an [AV2 bitstream](#) in the [Length delimited bitstream format](#) specified in Annex B of [AV2], or into any other carriage that does not itself delimit temporal units, a [Temporal Delimiter OBU](#) (framed with its `leb128()` length prefix) **SHALL** be prepended ahead of each sample's OBUs that does not already start with one.

§ 3.3.3. Sync sample

A sample is a sync sample if, for every [coded extended layer unit](#) present in the sample, the first coded frame OBU of that coded extended layer unit has `obu_type` equal to `OBU_CLOSED_LOOP_KEY`. Such a sample is a [Closed Random Access](#) point for the carried bitstream and corresponds to a SAP of type 1 or 2 as defined in Annex I of [ISOBMFF].

A sync sample **SHALL** be self-sufficient for starting decoding at that point for all layers: every OBU referenced by the coded frame OBUs of the sync sample **SHALL** be available either by being carried in the [AV2CodecConfigurationBox](#) of the active [AV2SampleEntry](#) or by being present in the sync sample itself, as required by the [availability of HLS OBUs](#) in [AV2].

NOTE: In ISOBMFF terms, sync samples are signalled using the SyncSampleBox ('[stss](#)') in unfragmented tracks, or by setting `sample_is_non_sync_sample` equal to 0 in fragmented tracks. When every sample in a track is a sync sample, the SyncSampleBox is omitted in unfragmented tracks (per [ISOBMFF]), and in fragmented tracks this is expressed through the default sample flags of the MovieExtendsBox ('[trex](#)') or TrackFragmentHeaderBox ('[tfhd](#)') with no per-sample override.

§ 4. CMAF AV2 track format

TBD

§ 5. Common Encryption

TBD

§ 6. Codecs Parameter String

TBD

§ Index

§ Terms defined by this specification

av02, in § 3.1	AV2CodecConfigurationBox, in § 3.2.2	AV2SampleEntry, in § 3.2.1
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§ Terms defined by reference

[AV2] defines the following terms:

AV2 bitstream
availability of HLS OBUs
Bridge Frame OBU
Closed Loop Key OBU
Closed Random Access
coded extended layer unit
Inter Frame
Intra frame
Key Frame
Layer Configuration Record OBU
Leading frame
Length delimited bitstream format
metadata group OBU syntax
Metadata OBU
metadata short OBU syntax
Multistream
OBU
obu_header_extension_flag
obu_mlayer_id
obu_tlayer_id
obu_type
obu_xlayer_id
Open Loop Key OBU
Open Random Access
ordering of OBUs in a temporal unit
Sequence Header OBU
show_existing_frame
Singlestream
Switch Frame
Switch OBU
Temporal Delimiter OBU
Temporal Unit

[ISOBMFF] defines the following terms:

btrt
cli
colr
ctts
iso6
mdcv
nclx
pasp
VisualSampleEntry

§ References

§ Normative References

[AV2]

AV2 Bitstream & Decoding Process Specification. Standard. URL: <https://av2.aomedia.org/v1.0.0/index.html>

[CMAF]

Information technology — Multimedia application format (MPEG-A) — Part 19: Common media application format (CMAF) for segmented media. Standard. URL: <https://www.iso.org/standard/71975.html>

[ISOBMFF]

Information technology — Coding of audio-visual objects — Part 12: ISO base media file format. Standard. URL: <https://www.iso.org/standard/83102.html>

[RFC2119]

Key words for use in RFCs to Indicate Requirement Levels. Standard. URL: <https://www.rfc-editor.org/info/rfc2119>

[RFC8174]

Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words. Standard. URL: <https://www.rfc-editor.org/info/rfc8174>